

Computer Science Internal

Assessment: Criterion E

Total Word Count: 415

Meeting The Success Criterion

After the final meeting with my client (See Appendix E-1), this program fulfils the client's desires and specified success criteria. Furthermore, other than altering the "Back" buttons to red and the "Continue/Moving Forward" buttons to green, the GUI of the program closely resembles the proposed GUI design approved by Mr.Brown.

As seen in the table below, each individual success criterion has been tested concerning the test plan in Criterion B, where each test has been successful and fulfils the client's desires and success criterion.

Test Number	Action to Test	Method of Testing	Expected Outcome	Success Criteria	Tested	Result
1	Login – Invalid Credentials + Validation Rules	Enter various combinations of incorrect username/email and/or password	System denies access and allows incorrect credentials to be handled	1.a + 1.c	✓	Yes
2	Login – Valid Student	Enter correct Student credentials	System verifies credential and provides access to Student Main Menu correlated to the Student credentials	1.b + 1.d	✓	Yes
3	Login – Valid Teacher	Enter correct Teacher credentials	System verifies credential and provides access to Teacher Main Menu correlated to the Teacher credentials	1.b + 1.e	✓	Yes

4	Student – Choose Question Language	Navigate to StudentQuestionBank GUI by clicking “Practice Coding” button	System provides both Java and Python QuestionBank GUIs and presents their associated language questions	2.a.i	✓	Yes
5	Recommended Questions Algorithm	Complete several questions (both correctly and incorrectly)	Verify the recommendation algorithm adapts to Student’s attempts in questions, providing different recommended questions when checked at different times	2.a.iii	✓	Yes
6	Student – Testing Student Code	Submit a different samples of student question answers (some correct, some partially correct, and some completely wrong)	The test case table’s results align with the respective submitted code	2.a.ii	✓	Yes
7	Progress Bar during Question	Submit partially correct code	The progress bars aligns with the number of successful test cases	2.a.iv	✓	Yes
8	Solution Hint Box	Make 3 failed attempts in a question	The hint appears only after the 3 failed attempts	2.a.v	✓	Yes
9	Teacher – Manage Questions	Click on the Manage Questions button and navigate through the Teacher interface to add, edit, and delete questions in the question’s dashboard	The added, edited, and deleted questions should be reflected within the CyberDuck database and in the questions table.	2.b.i	✓	Yes

10	Teacher – Manage Students	Click on the Manage Students button and navigate through the Teacher interface to edit and delete students in the students dashboard and click the “View Progress” button to view individual student progress	The edited and deleted questions should be reflected within the CyberDuck database and the students table, and individual student progress is shown after clicking the “View Progress” button by displaying their individual attempts of questions and number of correct and incorrect attempts	2.b.ii	✓	Yes
11	Teacher – Manage Classes	Click on the Manage Classes button and navigate through the Teacher interface to add, edit, and delete classes in the classes’ dashboard	The added, edited, and deleted classes should be reflected within the CyberDuck database and in the classes table.	2.b.iii	✓	Yes

Recommendations for Future Improvements:

Although teachers can add a student to the database, students and teachers are unable to sign up as new users to the program with their username and password. A SignUp_GUI could be created where the user can input their information, selecting whether they are a student or a teacher, along with subsequent necessary information, where the SignUp_GUI will run a method in the User class to create the Student/Teacher object and update the database with their information, allowing them to log into the program. This would enable the secure addition of new users through the program using

current front-end and back-end protection via encapsulation and the argon2 library to have the new user's password hashed within the database.

Furthermore, the data types possible for the question parameters are int, long, double, float, boolean, and String. Thus, another future improvement would be expanding the data types possible for the question parameter. Code to parse String into other data types could be added into the parseParametersString method in the Questions class, allowing any data type desired to be utilized in the program as a question parameter. Furthermore, to consider further data types for the return type, code to parse String into other data types can be added into the runCode method in the Questions class, specifically in the runPythonCode and runJavaCode methods where the output is processed and returned.

A final improvement would be a reset email/username and password feature, as currently, users are unable to. This can be implemented by adding a "Forgot Password" button in the Login_GUI class that opens a new ResetPassword_GUI. The ResetPassword_GUI would allow users to input their email address, which would call a new method in the User class to confirm their information is in the database, and when the user inputs a new password, the method will hash the new password with the argon2 library, use the password mutator to update the User Object password variable, and then update the database with the new hashed password.